

Blinking LED Patterns Using Arduino Nano

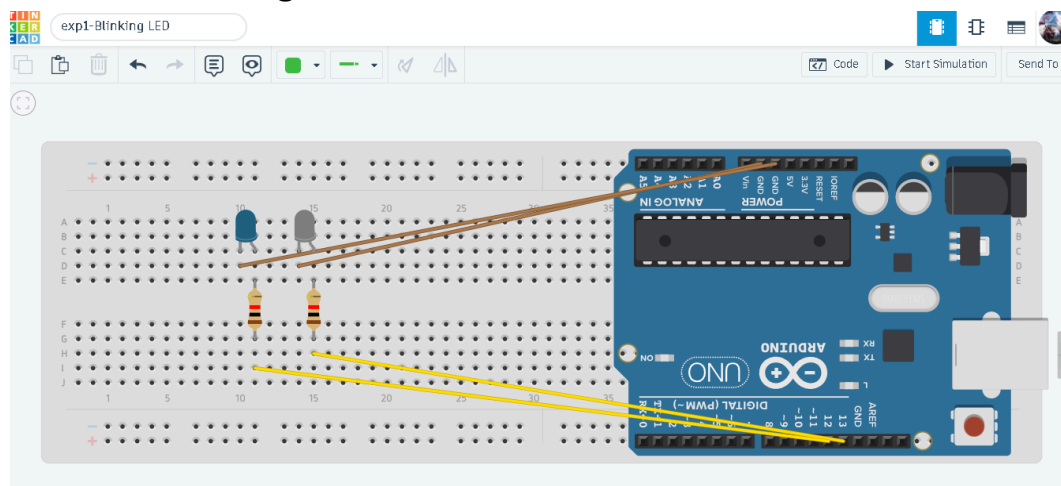
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• Components Required

Sr.No.	Component	Quantity	Description	Use
1.	Breadboard	1	A board with small holes used to make circuits.	Used to connect our components and build the circuit easily.
2.	Arduino Nano	1	A small microcontroller board used for programming electronic projects.	Used to control the LEDs by running our code.
3.	LED	2	A light that glows when current flows through it.	Used to display the blinking pattern using our code.
4.	Resistor	2	A resistor is used to reduce the flow of current.	Used to protect the LED from excessive current.
5.	Wire	4	Wires used to connect components together.	Used to make connections between the Arduino and the LED.

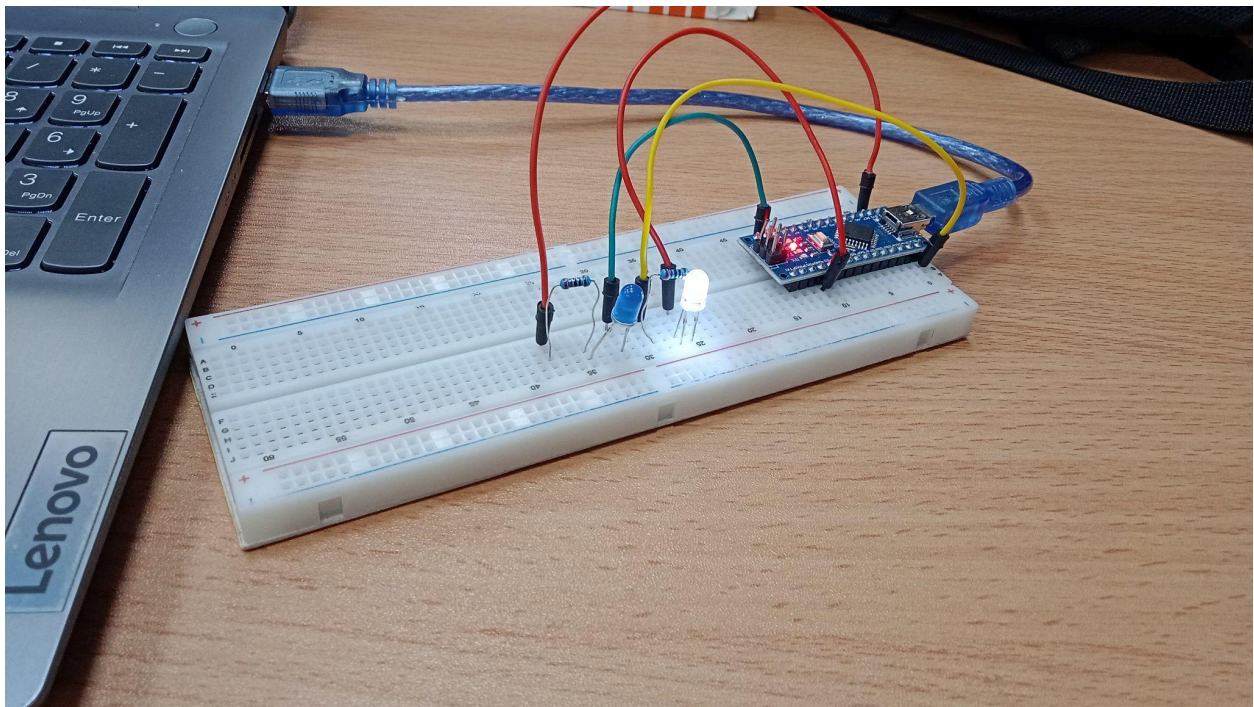
• Circuit Diagram

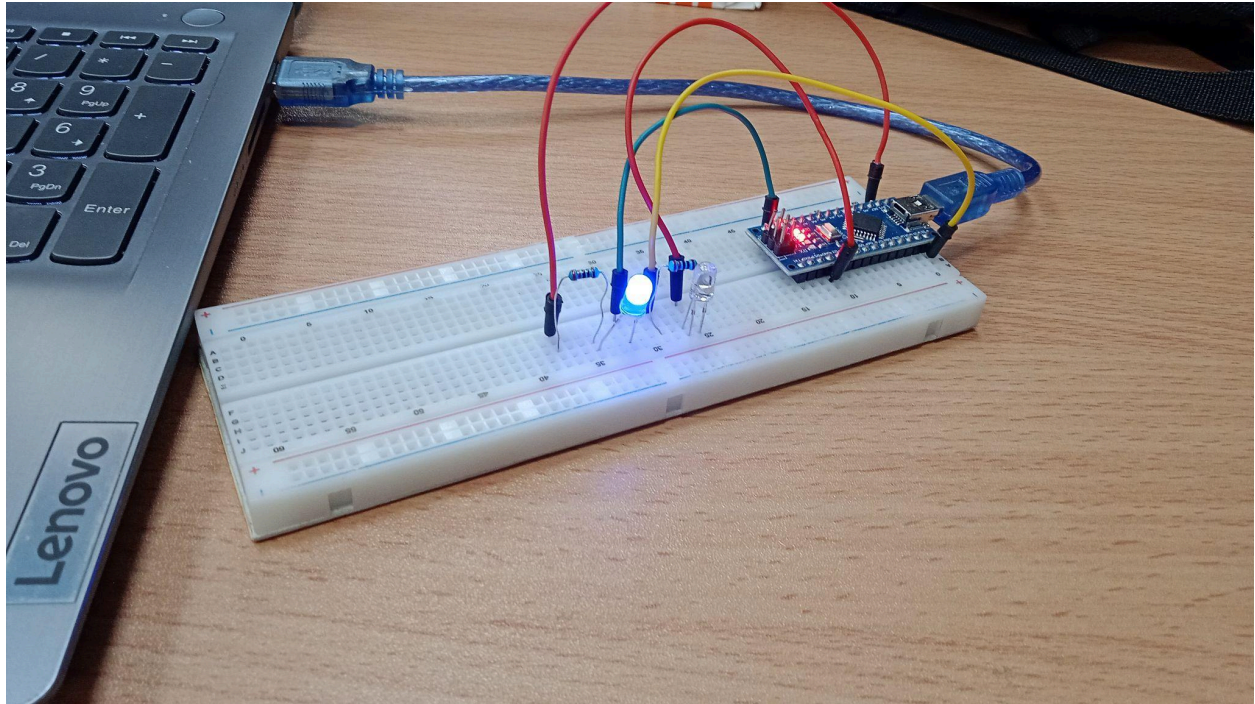


- **Program Code**

```
void setup() {  
  pinMode(LED_BUILTIN, OUTPUT);  
  pinMode(12, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(LED_BUILTIN, HIGH);  
  digitalWrite(12, LOW);  
  delay(500);  
  digitalWrite(LED_BUILTIN, LOW);  
  digitalWrite(12, HIGH);  
  delay(1000);  
}
```

- **Photograph of the Experiment**





- **Conclusion**

This experiment shows the basic working of the Arduino Nano and how to control LEDs using digital output pins. Used `pinMode()` ,`digitalWrite()` and `delay()` function to create different blinking timings of our circuit.